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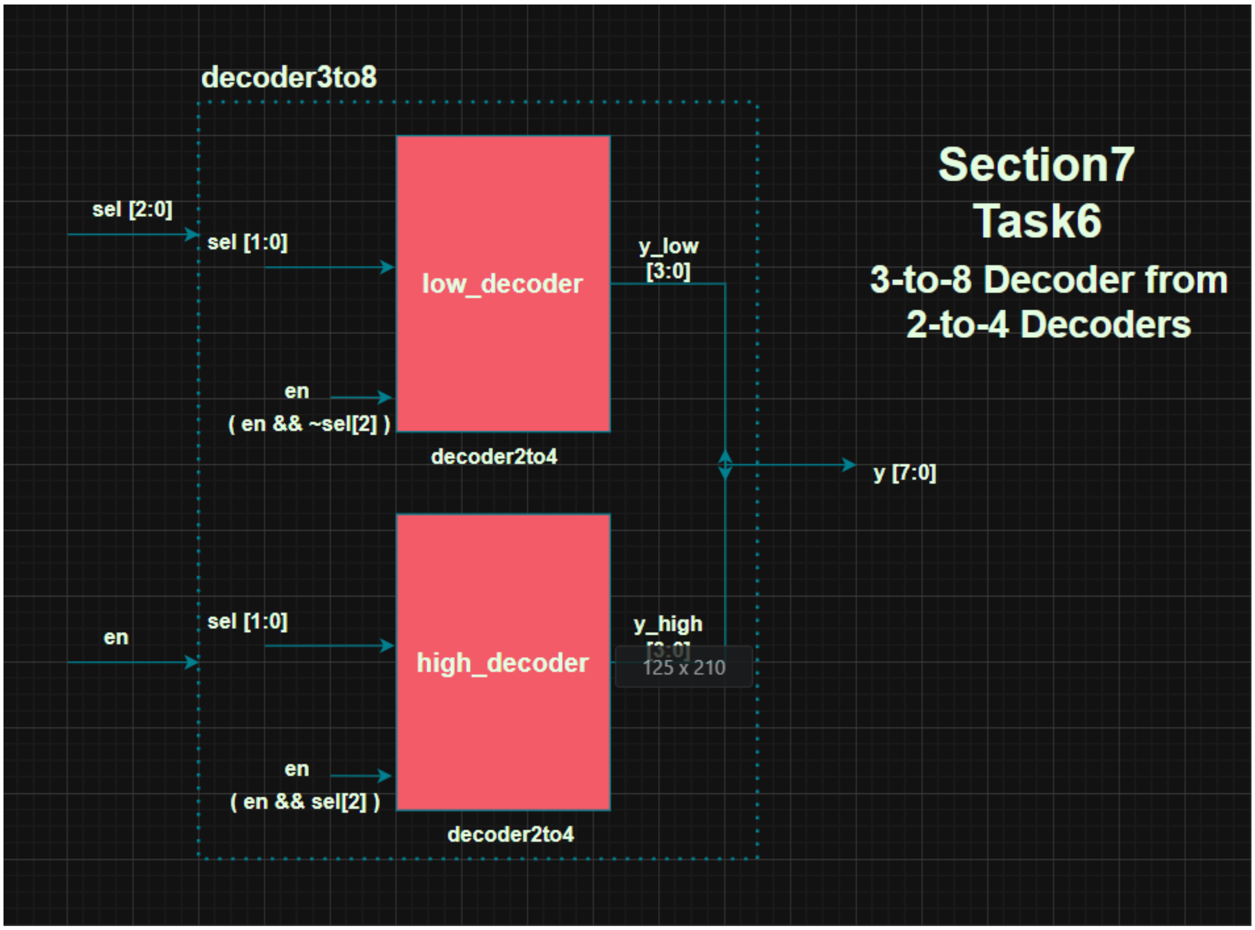
**1-EDA Playground link of Task6:** https://edaplayground.com/x/aiKQ

**EDA Playground link of Task7:** https://edaplayground.com/x/Scdc

**2-Schematic Diagram:**




**Task6_schematic_diagram.png**


**Task7_schematic_diagram.png**

### 3-Output Screenshot:

```
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 20 03:46 2026
[PASS] en=1 sel=0 (0) : y=1
[PASS] en=1 sel=1 (1) : y=10
[PASS] en=1 sel=10 (2) : y=100
[PASS] en=1 sel=11 (3) : y=1000
[PASS] en=0 sel=0 (0) : y=0
[PASS] en=0 sel=1 (1) : y=0
[PASS] en=0 sel=10 (2) : y=0
[PASS] en=0 sel=11 (3) : y=0
$finish called from file "testbench.sv", line 40.
$finish at simulation time                    90
           V C S   S i m u l a t i o n   R e p o r t
Time: 90 ns
CPU Time:      0.490 seconds;       Data structure size:   0.0Mb
Mon Jul 20 03:46:47 2026
Finding VCD file...
./dump.vcd
[2026-07-20 07:46:48 UTC] Opening EPWave...
Done
```

Task6_decoder2to4_log_result.png

```
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 20 03:45 2026
[PASS] en=1 sel=0 (0) : y=1
[PASS] en=1 sel=1 (1) : y=10
[PASS] en=1 sel=10 (2) : y=100
[PASS] en=1 sel=11 (3) : y=1000
[PASS] en=1 sel=100 (4) : y=10000
[PASS] en=1 sel=101 (5) : y=100000
[PASS] en=1 sel=110 (6) : y=1000000
[PASS] en=1 sel=111 (7) : y=10000000
[PASS] en=0 sel=0 (0) : y=0
[PASS] en=0 sel=1 (1) : y=0
[PASS] en=0 sel=10 (2) : y=0
[PASS] en=0 sel=11 (3) : y=0
[PASS] en=0 sel=100 (4) : y=0
[PASS] en=0 sel=101 (5) : y=0
[PASS] en=0 sel=110 (6) : y=0
[PASS] en=0 sel=111 (7) : y=0
$finish called from file "testbench.sv", line 40.
$finish at simulation time                  170
           V C S   S i m u l a t i o n   R e p o r t
Time: 170 ns
CPU Time:      0.560 seconds;       Data structure size:   0.0Mb
Mon Jul 20 03:45:23 2026
Finding VCD file...
./dump.vcd
[2026-07-20 07:45:23 UTC] Opening EPWave...
Done
```

Task6_log_result.png

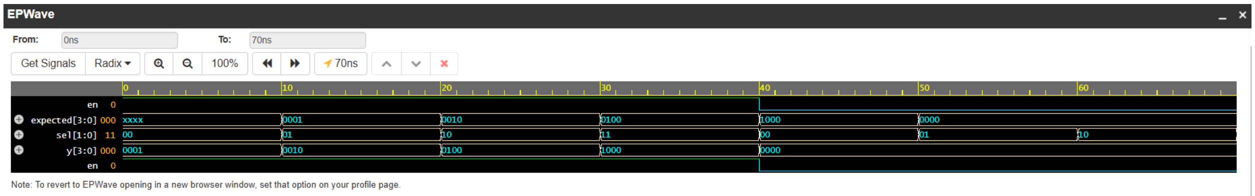
```
Log   Share
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 20 06:18 2026
[PASS] in=0 zeros=32
[PASS] in=1 zeros=31
[PASS] in=8000 zeros=16
[PASS] in=80000000 zeros=0
[PASS] in=5e23536 zeros=5
[PASS] in=57dea13c zeros=1
[PASS] in=be7fe77d zeros=0
[PASS] in=30aa33e2 zeros=2
[PASS] in=279440b zeros=6
[PASS] in=b9b381df zeros=0
[PASS] in=f350a940 zeros=0
[PASS] in=d51778f7 zeros=0
[PASS] in=6d7e79a6 zeros=1
[PASS] in=30d4b41b zeros=2
[PASS] in=25b9db5 zeros=6
[PASS] in=82e394fa zeros=0
[PASS] in=19bc7a4e zeros=3
[PASS] in=5a98a215 zeros=1
[PASS] in=7985c17d zeros=1
[PASS] in=e43a5f72 zeros=0
[PASS] in=ffbd4196 zeros=0
[PASS] in=217cb731 zeros=2
[PASS] in=9df554c4 zeros=0
[PASS] in=cac8ccaa zeros=0
[PASS] in=f58bc0c4 zeros=0
[PASS] in=11ab26cf zeros=3
[PASS] in=38646d4f zeros=2
[PASS] in=e8478af4 zeros=0
[PASS] in=7337dd17 zeros=1
[PASS] in=76264a88 zeros=1
[PASS] in=2f9fc9f1 zeros=2
[PASS] in=2da1e52c zeros=2
[PASS] in=94d34dce zeros=0
[PASS] in=9bee7505 zeros=0
[PASS] in=93fee0cb zeros=0
[PASS] in=3af0db8c zeros=2
[PASS] in=ac13f71a zeros=0
[PASS] in=8ef14f37 zeros=0
[PASS] in=4c23c660 zeros=1
```

Task7_log_result1.png

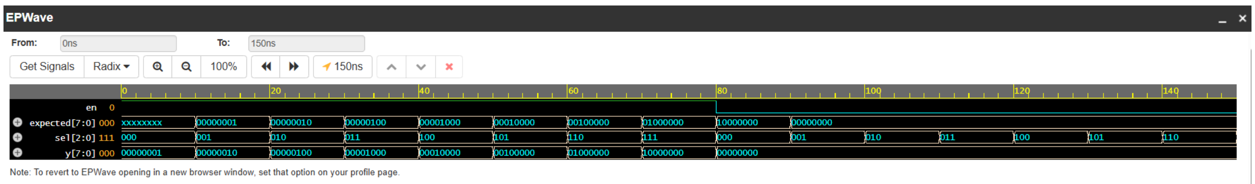
```
Log   Share
[PASS] in=9bee7505 zeros=0
[PASS] in=93fee0cb zeros=0
[PASS] in=3af0db8c zeros=2
[PASS] in=ac13f71a zeros=0
[PASS] in=8ef14f37 zeros=0
[PASS] in=4c23c660 zeros=1
[PASS] in=1ac5a95f zeros=3
[PASS] in=bb91ab7a zeros=0
[PASS] in=b1487ba2 zeros=0
[PASS] in=8a4b2cf0 zeros=0
[PASS] in=f7fb42c9 zeros=0
[PASS] in=7dc155dc zeros=1
[PASS] in=4a089e41 zeros=1
[PASS] in=84285d3f zeros=0
[PASS] in=2890cc12 zeros=2
[PASS] in=cded48f4 zeros=0
[PASS] in=a8be83df zeros=0
[PASS] in=53a24ec5 zeros=1
[PASS] in=cf8028d7 zeros=0
[PASS] in=92d2a694 zeros=0
[PASS] in=a5568088 zeros=0
[PASS] in=c4f6001 zeros=4
[PASS] in=867b8d31 zeros=0
[PASS] in=26b88d29 zeros=2
[PASS] in=65a127d6 zeros=1
[PASS] in=14ecc7f4 zeros=3
[PASS] in=c20e5ad9 zeros=0
[PASS] in=3d77c14f zeros=2
[PASS] in=fc69c200 zeros=0
[PASS] in=fe1f66dd zeros=0
[PASS] in=13f29dd2 zeros=3
All tests passed!
$finish called from file "testbench.sv", line 55.
$finish at simulation time             64000
           V C S   S i m u l a t i o n   R e p o r t
Time: 64000 ps
CPU Time:      0.570 seconds;       Data structure size:   0.0Mb
Mon Jul 20 06:18:46 2026
Finding VCD file...
./dump.vcd
[2026-07-20 10:18:47 UTC] Opening EPWave...
Done
```

Task7_log_result2.png

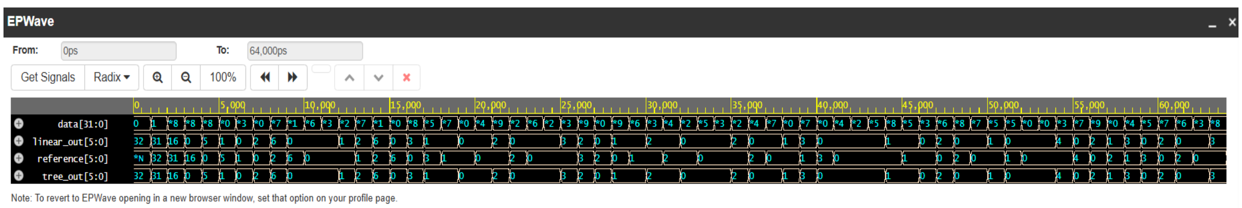
## 4-Waveform Screenshot:




Task6_decoder2to4_waveform.png




Task6_waveform.png




Task7_waveform.png

## 5-Explanation:

### Task7 Explanation:

The linear version has 32 logic levels in it's critical path because it checks each bit until he gets the 1, but for critical path we assume all 32 bits to be 0, whereas the tree version has exactly 5 logic levels because it divides the 32 bit in such a way that base case/base level would be 2-bit group and from that we go to 4-bit, than 8-bit, than 16-bit and after that finally 32-bit level (which is $5^{th}$ level from bottom to highest) after merging in each level.
Tree version is faster, from linear version, for large bit widths (N) because for large bit widths linear version grows O(N) whereas tree version grows logarithmically i.e. (log2N), which makes tree version significantly faster for large bit widths than linear version.